

Control Variates for Bayesian Estimation of a GARCH Model

TD project

May 4, 2026

Abstract

This note implements a random-walk Metropolis sampler for a Bayesian GARCH(1,1) model and compares plain MCMC estimates with zero-variance control variates. The first controls are the linear-polynomial controls of Mira, Solgi and Imparato (2013). Larger polynomial dictionaries are then handled either by ordinary least squares or by a Lasso screening step inspired by Leluc, Portier and Segers (2021). Numerical error is assessed by independent repeated runs and by batch-means standard errors.

1 Model and Data

For returns r_t , the model is

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \quad h_t = \omega_1 + \omega_2 r_{t-1}^2 + \omega_3 h_{t-1}.$$

The parameter vector is $\omega = (\omega_1, \omega_2, \omega_3)$. We use independent centered normal priors on ω_i , truncated to

$$\omega_1 > 0, \quad \omega_2 > 0, \quad \omega_3 > 0, \quad \omega_2 + \omega_3 < 0.995.$$

The likelihood is the usual Gaussian quasi-likelihood, up to constants,

$$\ell(\omega) = -\frac{1}{2} \sum_t \left\{ \log h_t + \frac{r_t^2}{h_t} \right\}.$$

Two datasets are used. The first is simulated from

$$\omega^\star = (0.04, 0.08, 0.88)$$

with 600 observations. The second is daily EUR/USD data from 2023-01-01 to 2024-12-31, converted to percentage log-returns. The exchange-rate data were downloaded from Frankfurter and saved locally in the project folder.

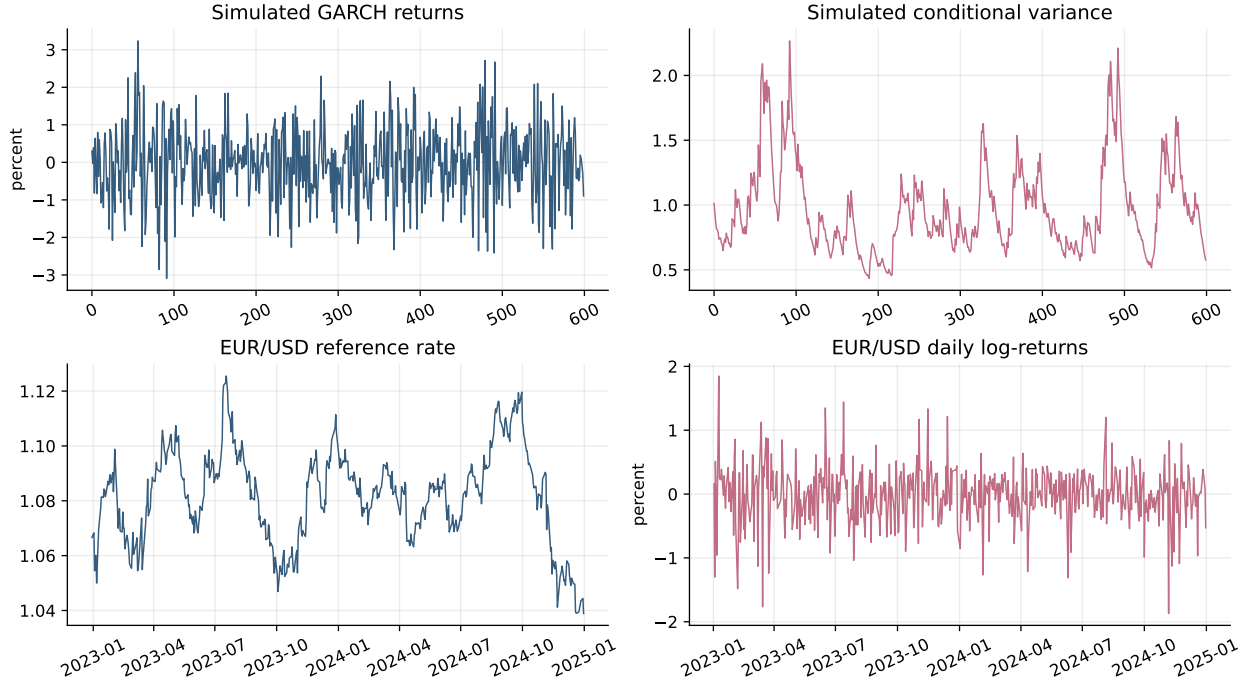


Figure 1: Simulated GARCH series and EUR/USD exchange-rate data.

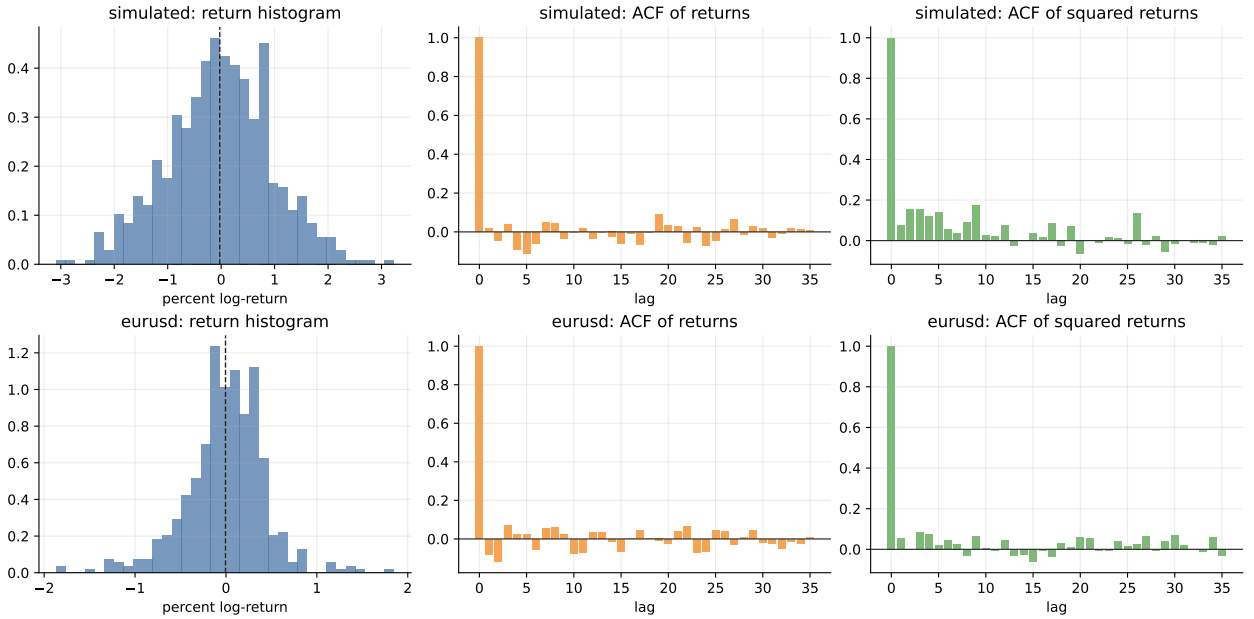


Figure 2: Extra data diagnostics: return histograms and autocorrelations of returns and squared returns.

2 Random-Walk Metropolis

The sampler is a random-walk Metropolis algorithm. To avoid negative proposals and to make the target support unconstrained, the chain is run on $\theta \in \mathbb{R}^3$, mapped to ω by

$$\omega_1 = e^{\theta_1}, \quad \omega_2 = \rho \frac{e^{\theta_2}}{1 + e^{\theta_2} + e^{\theta_3}}, \quad \omega_3 = \rho \frac{e^{\theta_3}}{1 + e^{\theta_2} + e^{\theta_3}}, \quad \rho = 0.995.$$

The transformed target includes the Jacobian of this map. The proposal is

$$\theta' = \theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Sigma_q).$$

We first tune a diagonal proposal by short pilot runs, then estimate the covariance of θ from a pilot chain and set Σ_q proportional to that empirical covariance. This reuses the TD Metropolis-Hastings idea of choosing a proposal variance adapted to the target shape. Each numerical experiment used 10 independent chains, 8000 Metropolis iterations per chain, burn-in 2000, and thinning by 3. This is longer than the first draft because short trace plots may show local upward or downward excursions. Those excursions are normal for a correlated random-walk chain; the more relevant checks are the running mean, autocorrelation, and repeated independent chains.

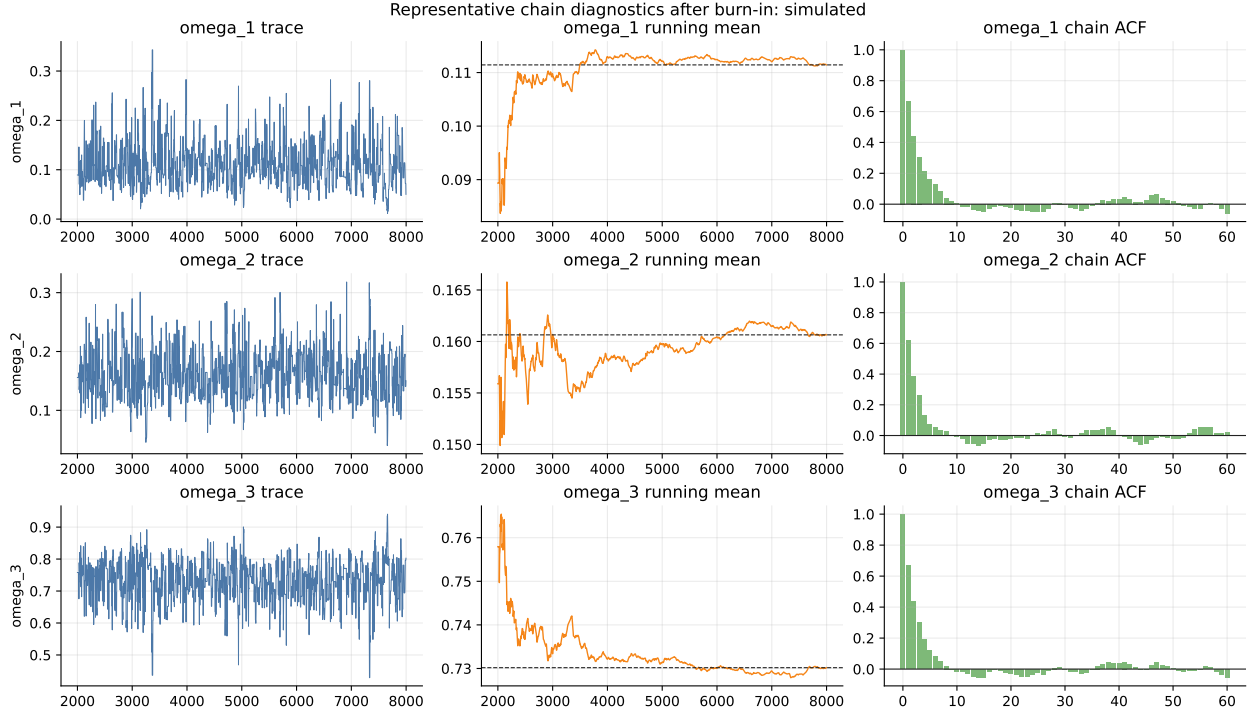


Figure 3: Representative chain diagnostics for the simulated dataset.

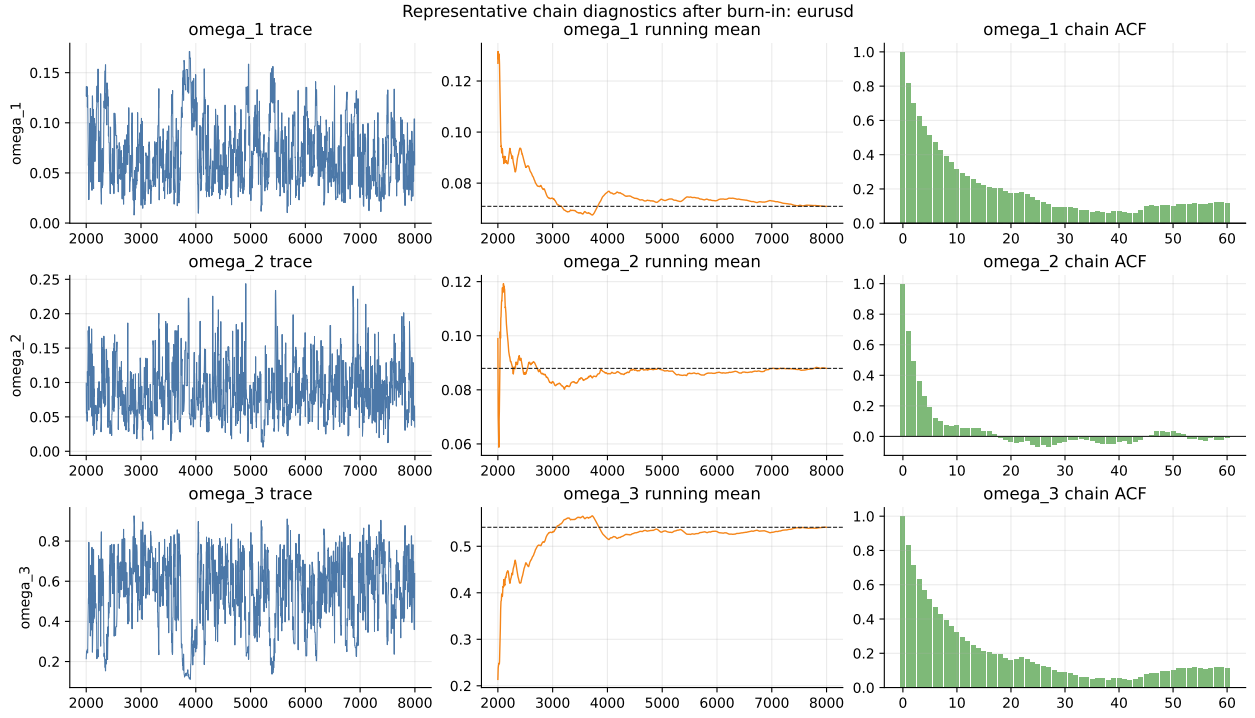


Figure 4: Representative chain diagnostics for EUR/USD.

3 Zero-Variance Control Variates

The control variates require the score of the posterior. For the likelihood part, set $D_{t,k} = \partial h_t / \partial \omega_k$. With fixed initial variance h_1 , $D_{1,k} = 0$, and for $t \geq 2$

$$D_{t,1} = 1 + \omega_3 D_{t-1,1}, \quad D_{t,2} = r_{t-1}^2 + \omega_3 D_{t-1,2}, \quad D_{t,3} = h_{t-1} + \omega_3 D_{t-1,3}.$$

Since

$$\ell(\omega) = -\frac{1}{2} \sum_{t=1}^T \left\{ \log h_t + \frac{r_t^2}{h_t} \right\},$$

the gradient of the log-likelihood is

$$\frac{\partial \ell}{\partial \omega_k} = \frac{1}{2} \sum_{t=1}^T \left(\frac{r_t^2}{h_t^2} - \frac{1}{h_t} \right) D_{t,k}.$$

The prior adds $-\omega_k / \sigma_k^2$. The sampler is implemented in θ , so the code applies

$$\nabla_{\theta} \log \pi_{\theta}(\theta) = J_{\omega}(\theta)^{\top} \nabla_{\omega} \log \pi_{\omega}(\omega(\theta)) + \nabla_{\theta} \log |\det J_{\omega}(\theta)|.$$

For a target density $\pi(x)$, Mira, Solgi and Imparato define

$$z(x) = -\frac{1}{2} \nabla \log \pi(x).$$

For a polynomial trial function P , the controlled integrand is

$$\tilde{f}(x) = f(x) - \frac{1}{2} \Delta P(x) + \nabla P(x)^{\top} z(x).$$

If $P(x) = a^{\top} x$, then $\Delta P = 0$ and

$$\tilde{f}(x) = f(x) + a^{\top} z(x).$$

The optimal a is obtained by minimizing the empirical variance. Equivalently, regress $f(x_i)$ on the control vector $z(x_i)$; if the regression coefficient is $\hat{\beta}$, the controlled estimate is

$$\hat{\mu}_{cv} = \bar{f} - \bar{z}^\top \hat{\beta}.$$

In this project $x = \theta$, while $f_j(\theta) = \omega_j(\theta)$, because the posterior summaries are desired in the original GARCH parameters.

4 Larger Dictionaries and Lasso

For all monomials up to total degree q in dimension $d = 3$, the number of non-constant polynomial controls is

$$\binom{d+q}{d} - 1.$$

This grows quickly: degree 1 gives 3 controls, degree 2 gives 9, and degree 5 gives 55. Ordinary least squares on all controls can become unstable because the design matrix becomes highly collinear; it also becomes expensive when d or q is larger.

Following Leluc, Portier and Segers, we use Lasso as a screening step. The implementation standardizes the controls, computes a Lasso path by coordinate descent, selects the penalty by BIC, and finally refits ordinary least squares only on the selected controls. This is the “ZV-Lasso-5” method in the tables.

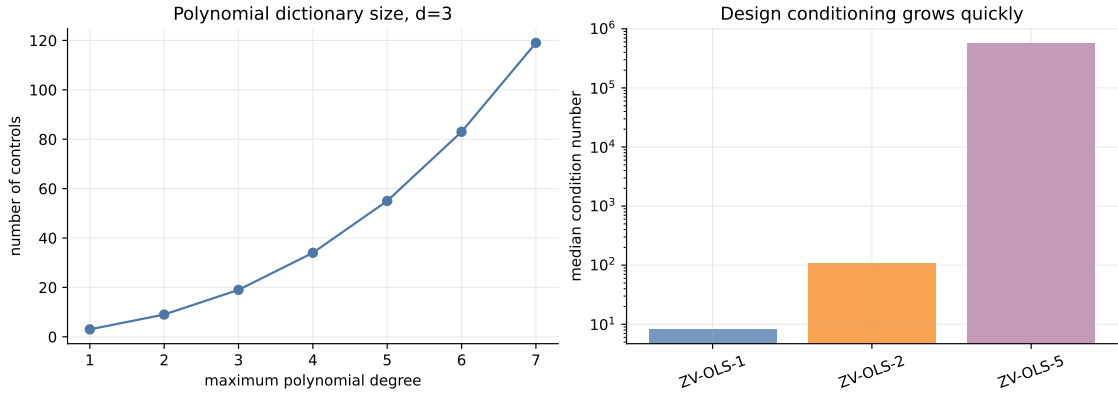


Figure 5: Number of polynomial controls and conditioning of the OLS design.

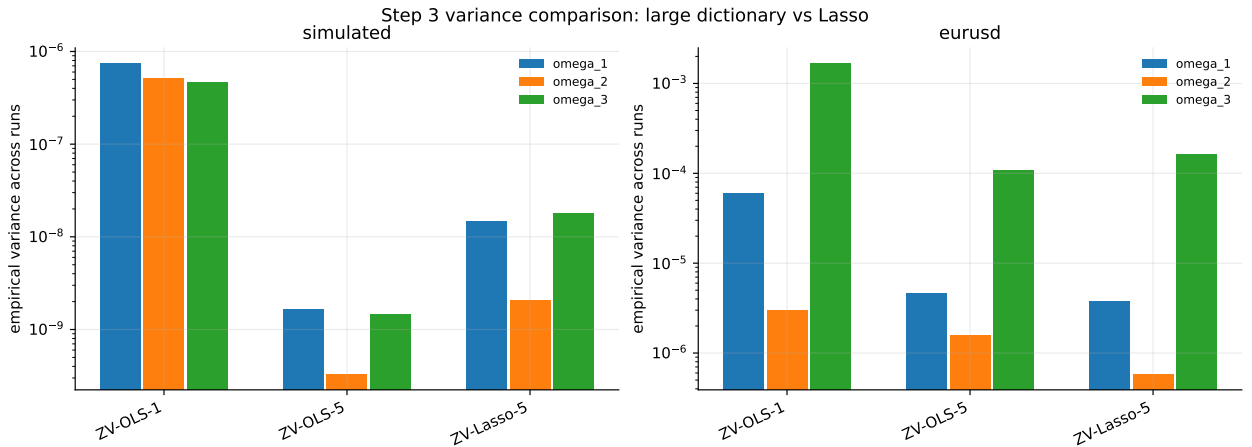


Figure 6: Step 3 variance comparison. The figure compares first-order controls, full degree-5 OLS, and degree-5 Lasso screening.

5 Numerical Results

The boxplots show the repeated-run variability of each posterior mean estimate. For the simulated case, the dashed line is the data-generating value; finite-sample posterior means need not equal it. For EUR/USD, the dashed line is the pooled raw-MC mean, used only as a visual reference.

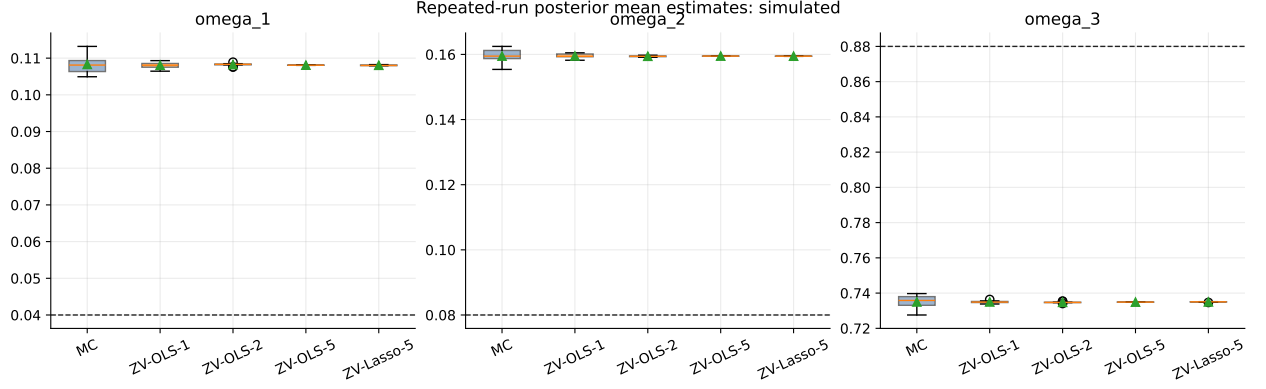


Figure 7: Repeated-run estimates for simulated GARCH data.

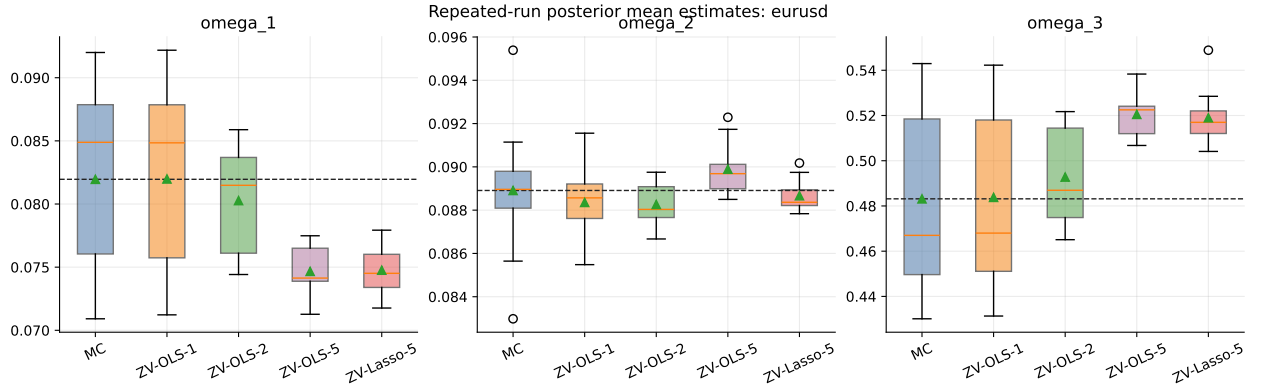


Figure 8: Repeated-run estimates for EUR/USD log-returns.

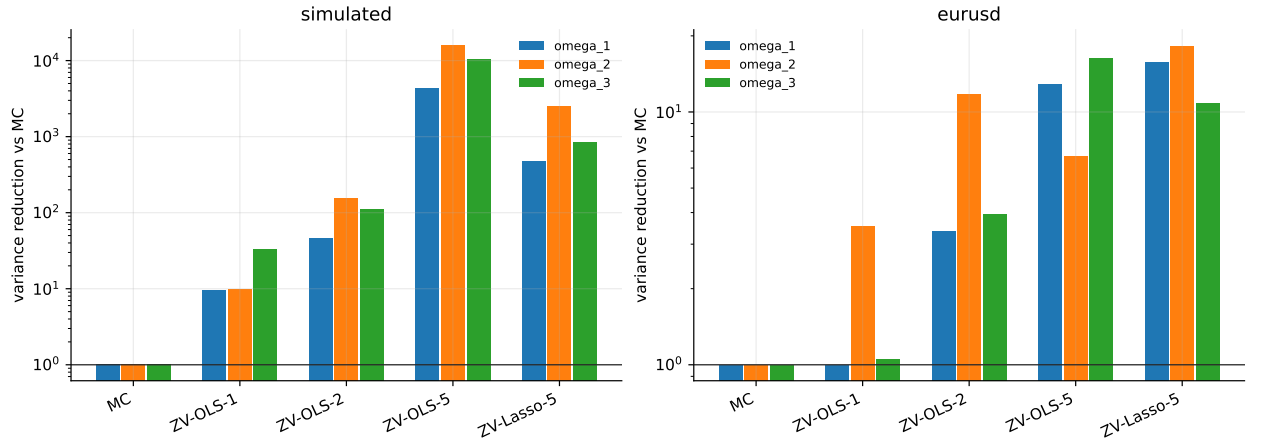


Figure 9: Empirical variance reduction relative to raw Monte Carlo. Values above 1 improve over MC.



Figure 10: Variance comparison for the first-order control-variate step only.

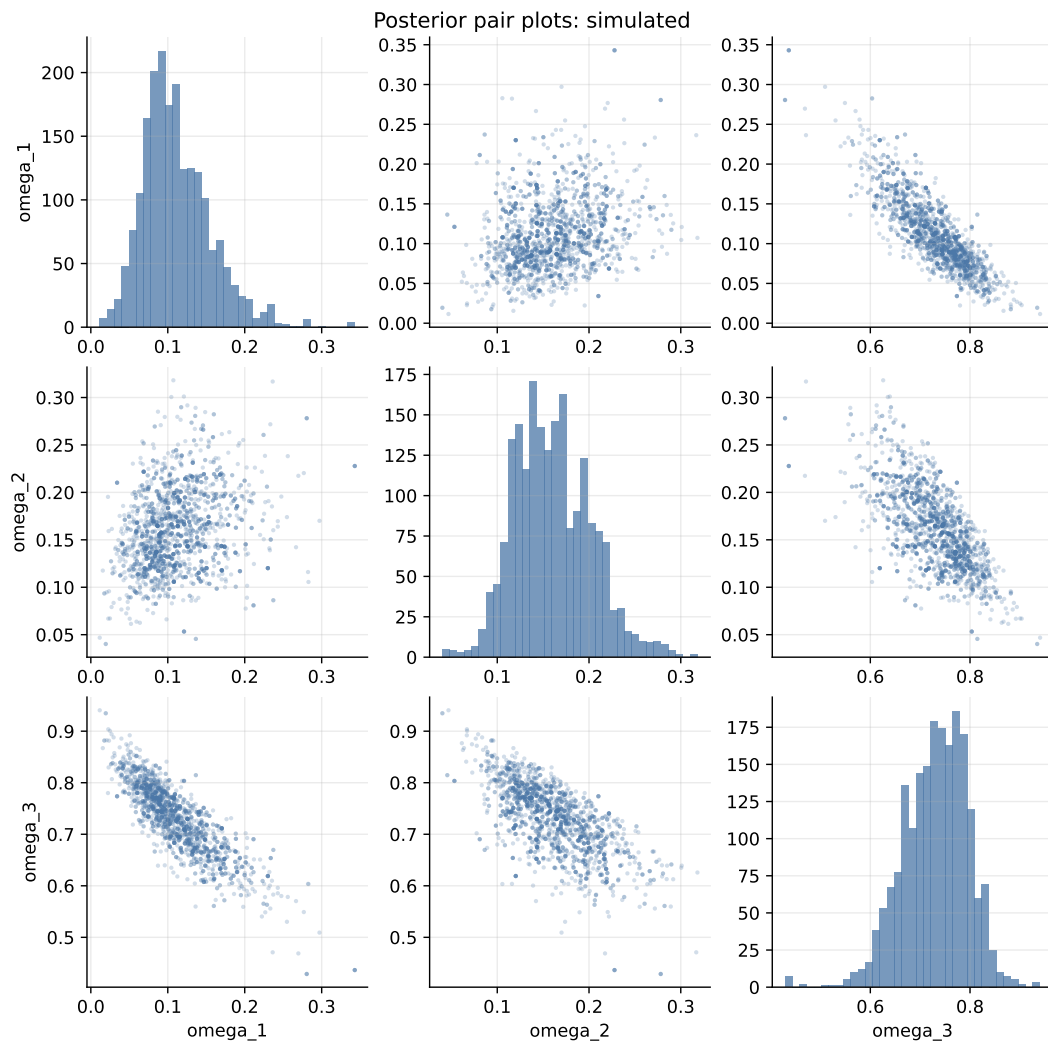


Figure 11: Posterior pair plot for the simulated dataset.

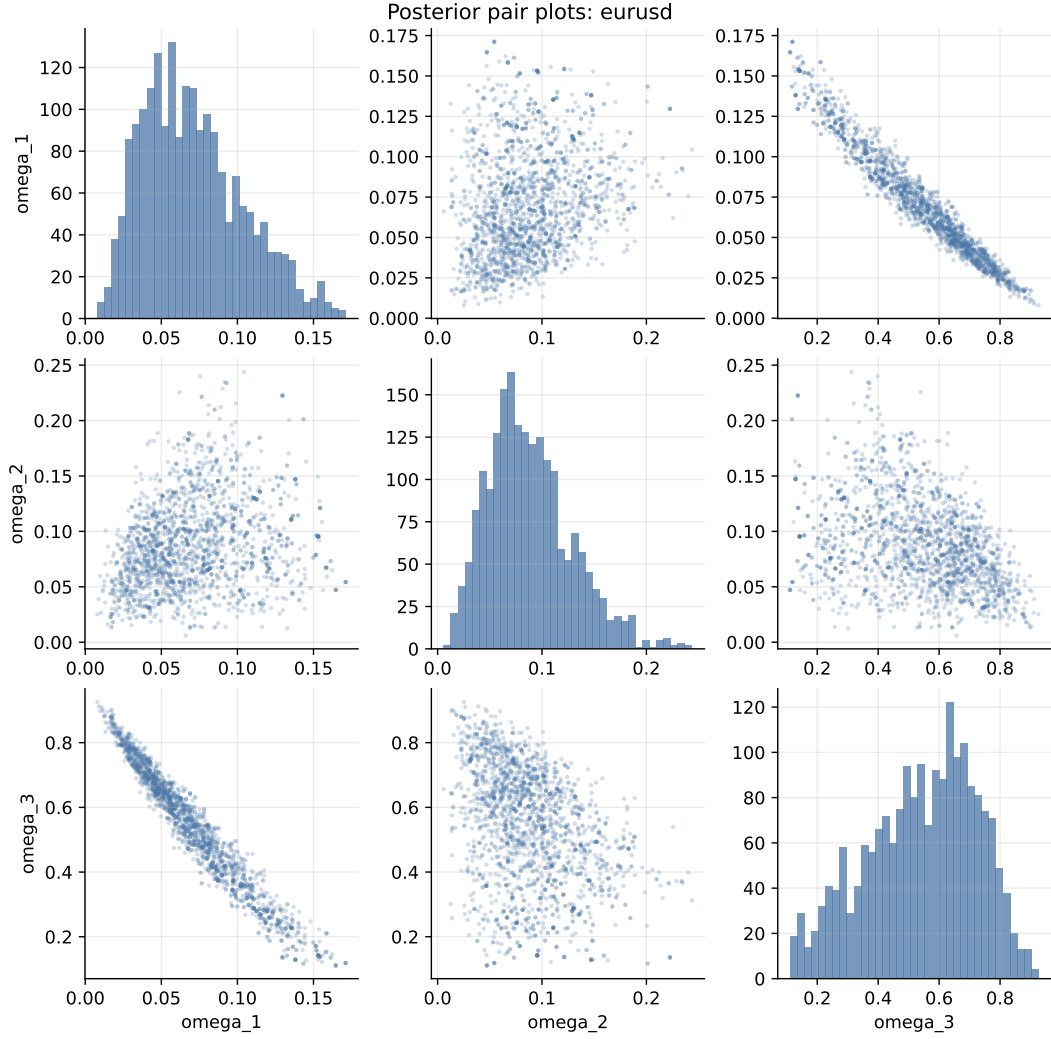


Figure 12: Posterior pair plot for EUR/USD.

5.1 Simulated Data

data	param.	method	mean	sd across runs	rmse/ref.	var. red.	avg. selected
simulated	omega_1	MC	0.1083	0.0026	0.0683	1.0000	0.0000
simulated	omega_1	ZV-Lasso-5	0.1081	1.21e-04	0.0681	475.1787	27.0000
simulated	omega_1	ZV-OLS-1	0.1081	8.62e-04	0.0681	9.4103	3.0000
simulated	omega_1	ZV-OLS-2	0.1082	3.89e-04	0.0682	46.1652	9.0000
simulated	omega_1	ZV-OLS-5	0.1081	4.05e-05	0.0681	4.26e+03	55.0000
simulated	omega_2	MC	0.1595	0.0023	0.0795	1.0000	0.0000
simulated	omega_2	ZV-Lasso-5	0.1595	4.54e-05	0.0795	2.50e+03	19.2000
simulated	omega_2	ZV-OLS-1	0.1595	7.21e-04	0.0795	9.9389	3.0000
simulated	omega_2	ZV-OLS-2	0.1594	1.84e-04	0.0794	152.2873	9.0000
simulated	omega_2	ZV-OLS-5	0.1595	1.81e-05	0.0795	1.57e+04	55.0000
simulated	omega_3	MC	0.7350	0.0039	0.1451	1.0000	0.0000
simulated	omega_3	ZV-Lasso-5	0.7350	1.35e-04	0.1450	838.8597	22.1000
simulated	omega_3	ZV-OLS-1	0.7350	6.85e-04	0.1450	32.4641	3.0000
simulated	omega_3	ZV-OLS-2	0.7348	3.70e-04	0.1452	111.1825	9.0000
simulated	omega_3	ZV-OLS-5	0.7349	3.82e-05	0.1451	1.05e+04	55.0000

Table 1: Repeated-run summary for simulated data. “sd across runs” is the empirical numerical error across the 10 independent chains.

The first-order controls already reduce variance substantially. Degree-2 controls improve further. Degree-5 OLS is extremely effective on this low-dimensional simulated example, but this is also the setting where it is easiest for OLS to exploit the full dictionary without numerical problems. Post-Lasso keeps fewer controls and still gives a large variance reduction.

5.2 EUR/USD Data

data	param.	method	mean	sd across runs	rmse/ref.	var. red.	avg. selected
eurusd	omega_1	MC	0.0820	0.0077	0.0073	1.0000	0.0000
eurusd	omega_1	ZV-Lasso-5	0.0748	0.0019	0.0074	15.7828	41.2000
eurusd	omega_1	ZV-OLS-1	0.0820	0.0077	0.0073	1.0010	3.0000
eurusd	omega_1	ZV-OLS-2	0.0803	0.0042	0.0043	3.3796	9.0000
eurusd	omega_1	ZV-OLS-5	0.0747	0.0021	0.0076	12.9199	55.0000
eurusd	omega_2	MC	0.0889	0.0033	0.0031	1.0000	0.0000
eurusd	omega_2	ZV-Lasso-5	0.0887	7.62e-04	7.62e-04	18.2930	30.0000
eurusd	omega_2	ZV-OLS-1	0.0884	0.0017	0.0017	3.5313	3.0000
eurusd	omega_2	ZV-OLS-2	0.0883	9.51e-04	0.0011	11.7247	9.0000
eurusd	omega_2	ZV-OLS-5	0.0899	0.0013	0.0015	6.7051	55.0000
eurusd	omega_3	MC	0.4832	0.0420	0.0399	1.0000	0.0000
eurusd	omega_3	ZV-Lasso-5	0.5190	0.0128	0.0379	10.8039	40.6000
eurusd	omega_3	ZV-OLS-1	0.4838	0.0409	0.0388	1.0582	3.0000
eurusd	omega_3	ZV-OLS-2	0.4928	0.0212	0.0223	3.9454	9.0000
eurusd	omega_3	ZV-OLS-5	0.5205	0.0104	0.0386	16.3123	55.0000

Table 2: Repeated-run summary for EUR/USD returns. The reference for the RMSE column is the pooled raw-MC mean.

For the real-data example the full degree-5 OLS estimator is less reliable: for ω_1 and ω_3 its repeated-run variance is worse than MC. This matches the conditioning issue: a larger dictionary can help, but the naive OLS fit may become unstable. Degree-2 OLS is robust here, and the Lasso-screened estimator improves ω_2 while avoiding the full degree-5 regression.

5.3 Focused Variance Tables

dataset	parameter	method	empirical_variance	variance_reduction
eurusd	omega_1	MC	5.96e-05	1.0000
eurusd	omega_1	ZV-OLS-1	5.95e-05	1.0010
eurusd	omega_2	MC	1.06e-05	1.0000
eurusd	omega_2	ZV-OLS-1	3.00e-06	3.5313
eurusd	omega_3	MC	0.0018	1.0000
eurusd	omega_3	ZV-OLS-1	0.0017	1.0582
simulated	omega_1	MC	7.00e-06	1.0000
simulated	omega_1	ZV-OLS-1	7.43e-07	9.4103
simulated	omega_2	MC	5.16e-06	1.0000
simulated	omega_2	ZV-OLS-1	5.20e-07	9.9389
simulated	omega_3	MC	1.52e-05	1.0000
simulated	omega_3	ZV-OLS-1	4.69e-07	32.4641

Table 3: Variance comparison for the first-order control-variate step only.

dataset	parameter	method	empirical_variance	variance_reduction	mean_selected	median_condition
eurusd	omega_1	ZV-Lasso-5	3.78e-06	15.7828	41.2000	2.62e+05
eurusd	omega_1	ZV-OLS-1	5.95e-05	1.0010	3.0000	8.5186
eurusd	omega_1	ZV-OLS-5	4.61e-06	12.9199	55.0000	1.40e+07
eurusd	omega_2	ZV-Lasso-5	5.80e-07	18.2930	30.0000	2.77e+04
eurusd	omega_2	ZV-OLS-1	3.00e-06	3.5313	3.0000	8.5186
eurusd	omega_2	ZV-OLS-5	1.58e-06	6.7051	55.0000	1.40e+07
eurusd	omega_3	ZV-Lasso-5	1.64e-04	10.8039	40.6000	2.26e+05
eurusd	omega_3	ZV-OLS-1	0.0017	1.0582	3.0000	8.5186
eurusd	omega_3	ZV-OLS-5	1.08e-04	16.3123	55.0000	1.40e+07
simulated	omega_1	ZV-Lasso-5	1.47e-08	475.1787	27.0000	3.51e+04
simulated	omega_1	ZV-OLS-1	7.43e-07	9.4103	3.0000	8.1977
simulated	omega_1	ZV-OLS-5	1.64e-09	4.26e+03	55.0000	5.78e+05
simulated	omega_2	ZV-Lasso-5	2.06e-09	2.50e+03	19.2000	5.68e+03
simulated	omega_2	ZV-OLS-1	5.20e-07	9.9389	3.0000	8.1977
simulated	omega_2	ZV-OLS-5	3.29e-10	1.57e+04	55.0000	5.78e+05
simulated	omega_3	ZV-Lasso-5	1.82e-08	838.8597	22.1000	1.95e+04
simulated	omega_3	ZV-OLS-1	4.69e-07	32.4641	3.0000	8.1977
simulated	omega_3	ZV-OLS-5	1.46e-09	1.05e+04	55.0000	5.78e+05

Table 4: Step 3 variance comparison: full degree-5 OLS versus Lasso-screened degree-5 controls.

6 MCMC Dependence and Regression

The linear regression used to estimate control-variate coefficients is not literally an iid regression because the inputs are MCMC draws. The coefficient estimate can still be consistent under ergodic-average conditions, but the usual iid regression standard errors are not valid. We addressed this in two ways:

- numerical error is reported through repeated independent chains, which directly assesses estimator variability;
- batch-means standard errors are computed in the notebook for the controlled output series.

Other practical solutions are to estimate coefficients on a pilot chain and evaluate on a second independent chain, to fit on thinned samples, or to use block cross-validation for Lasso.

parameter	full_sample_estimate	full_batch_se	thin5_estimate	thin5_batch_se
omega_1	0.108072	0.000011	0.107956	0.000007
omega_2	0.159524	0.000002	0.159462	0.000002
omega_3	0.734892	0.000011	0.735033	0.000008

Table 5: A small thinning check on the simulated representative chain. Thinning changes little here, suggesting that the regression fit is not dominated by adjacent highly dependent draws.

7 Conclusion

The project confirms the main message of zero-variance control variates. Once $\nabla \log \pi$ is available, first-order polynomial controls are easy to add to an MCMC estimator and can sharply reduce numerical error. Higher-order controls can reduce the variance further, but the OLS regression can become ill-conditioned. Lasso screening is a practical compromise: it lets us start from a larger control dictionary without always paying the numerical price of the full regression.

References

- [1] A. Mira, R. Solgi, and D. Imparato. Zero variance Markov chain Monte Carlo for Bayesian estimators. *Statistics and Computing*, 23:653–662, 2013. doi:10.1007/s11222-012-9344-6.

- [2] R. Leluc, F. Portier, and J. Segers. Control variate selection for Monte Carlo integration. *Statistics and Computing*, 31:50, 2021. doi:10.1007/s11222-021-10011-z.
- [3] Frankfurter exchange-rate API. <https://www.frankfurter.app/docs/>.